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Human immunodeficiency virus prevalence, incidence, and residual transmission risk in first-time and repeat blood donations in Zimbabwe: implications on blood safety

Tonderai Mapako, David A. Mvere, McLeod E. Chitiyo, Simbarashe Rusakaniko, Maarten J. Postma, and Marinus van Hulst

BACKGROUND: National Blood Service Zimbabwe human immunodeficiency virus (HIV) risk management strategy includes screening and discarding of first-time donations, which are collected in blood packs without an anticoagulant (dry pack). To evaluate the impact of discarding first-time donations on blood safety the HIV prevalence, incidence, and residual risk in first-time and repeat donations (wet packs) were compared.

STUDY DESIGN AND METHODS: Donor data from 2002 to 2010 were retrieved from a centralized national electronic donor database and retrospectively analyzed. Chi-square test was used to compare HIV prevalence with relative risk (RR), and the RR point estimates and 95% confidence interval (CI) are reported. Trend analysis was done using Cochran-Armitage trend test. HIV residual risk estimates were determined using published residual risk estimation models.

RESULTS: Over the 9 years the overall HIV prevalence estimates are 1.29% (n = 116,058) and 0.42% (n = 434,695) for first-time and repeat donations, respectively. The overall RR was 3.1 (95% CI, 2.9-3.3; $p < 0.0001$). The overall mean residual transmission risk of HIV window phase donations in first-time was 1:7384 (range, 1:11,308-1:5356) and in repeat donors it was 1:5496 (range, 1:9943-1:3347).

CONCLUSION: The significantly high HIV prevalence estimates recorded in first-time over repeat donations is indicative of the effectiveness of the HIV risk management strategy. However, comparable residual transmission risk estimates in first-time and repeat donors point to the need to further review the risk management strategies. Given the potential wastage of valuable resources, future studies should focus on the cost-effectiveness and utility of screening and discarding first-time donations.

Zimbabwe is categorized as a country with generalized epidemic of human immunodeficiency virus (HIV). It lies at the epicenter of the HIV epidemic in the Southern African region and has experienced one of the world's most severe HIV and AIDS crises.[1,2] However, it has recorded a significant decline in HIV prevalence in the general population from 18.1% (95% confidence interval [CI], 16.9%-18.8%) in 2005 and 2006 to 15.2% (95% CI, 14.3%-16.1%) in 2010 and 2011 based on Zimbabwe Demographic and Health Surveys.[3,4] There are several publications that have documented the

ABBREVIATIONS: NBSZ = National Blood Service Zimbabwe; RR = relative risk; TTI(s) = transfusion-transmissible infection(s).

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HIV decline in Zimbabwe, which started in the late 1990s.[5-11] The population of Zimbabwe based on the 2002 census is 11.6 million with 99.3% being of African ethnic group and 55% in the age group 15 to 64 years. The life expectancy at birth is 45 years compared to 61 years in 1992.[12]

The National Blood Service Zimbabwe (NBSZ) on average collects approximately 80,000 units of whole blood annually based on the internationally endorsed principle of 100% voluntary nonremunerated blood donation.[13] Donor selection is based on locally determined risk factors and all donated blood is routinely tested in a quality assured manner for transfusion-transmissible infections (TTIs) using serologic markers for HIV, hepatitis B virus (HBV), hepatitis C virus, and syphilis. NBSZ instituted various HIV and other TTI risk management strategies including screening and discarding of first-time donations, which started in 1992.[14] First-time blood donors donate blood in a blood pack without an anticoagulant (dry packs). Collecting donations in dry packs from this group reduces wastage of more expensive wet packs (with anticoagulant). Blood collected in dry pack coagulates and this acts as an extra safety measure to prevent transfusion and use of first-time donations, which may carry a higher risk of HIV and other TTIs such as HBV, which accounts for much of the discarded blood in Zimbabwe. Discarding first-time donations has been discussed recently because of the potential wastage of blood donations in a country with intermittent blood shortages and a limited health budget. In this study, we evaluate the impact of discarding first-time donations in Zimbabwe on blood safety by comparing the HIV prevalence, incidences, and residual risks in first-time and repeat donations. This forms the basis for continued or alternative blood safety strategies that NBSZ will use to ensure and safeguard the sustainability, adequacy, and safety of blood and blood products supply in Zimbabwe.

MATERIALS AND METHODS

This was a retrospective study conducted among all Zimbabwe blood donors from 2002 to 2010. Details of the blood safety program in Zimbabwe are given below.

Blood safety

NBSZ from its inception has operated as an independent nonprofit company solely mandated by the Government of Zimbabwe through the Ministry of Health and Child Welfare to procure, test, process, and distribute blood and blood products to all health institutions in Zimbabwe. It is authorized and entrusted to ensure availability of adequate and safe blood and blood products in Zimbabwe. There is a National Blood Policy[15] and Standards for Blood Donation, Processing and Clinical Transfusion in Zimbabwe[16] that guides the blood safety program. The NBSZ Board through its medical scientific subcommittee continuously monitors and evaluates the blood safety activities in Zimbabwe. The NBSZ executive management is charged with responsibilities to implement standard operating procedures to ensure blood safety.

Donor recruitment is targeted at low-risk populations and updated risk factors are used to select blood donors as per the donor risk classification described under Materials and Methods. There are various blood donor mobilization programs targeting the youth such as Pledge 25[17] that was introduced in 1994 and other measures all aimed at promoting repeat donations in Zimbabwe. Intensive donor education is provided that is key to continued blood safety.

All donors are individually assessed for risk factors before blood donation through a structured donor interview process and review of available previous donation history and test results. If the previous donor test results are positive, the donor is prevented from making further donations. All donors with positive test results are informed and offered postdonation counseling through the NBSZ donor notification and counseling program, which also involves external counseling partners. The NBSZ view is that postdonation counseling is key to ensure that those donors with positive test results will be encouraged to self-defer from attempting to make future blood donations which might compromise blood safety.

Zimbabwe, through the NBSZ, was the third country in the world to introduce 100% routine enzyme-linked immunosorbent assay testing for HIV in donated blood in 1985.[18] This demonstrates the NBSZ commitment to ensure that the blood safety program in Zimbabwe is responsive to the emerging threats and strives to meet the international blood safety standards.

Donor risk classification

As part of a structured risk management strategy, NBSZ in 1994 developed blood donor risk categories based on r-codes, which are R1 (two or more previous donations within 12 months), R2 (one previous donation within 12 months), R3 (previous donation not within 12 months—lapsed donor), and R4 (new donor). Consistent with World Health Organization (WHO) recommendations on blood safety, repeat donations (R1 and R2) are advocated. R1 and R2 donations are expected to show a low HIV prevalence compared to R3 and R4. The lapsed donors are classified separately as these donations carry an elevated HIV transmission risk compared to R1 and R2. Different policies that are based on age and donation venue are applied to new donors (R4) although they are maintained as category. New donors aged 16 to 20 years donating at a school venue are generally deemed to be safe[18] and their blood is collected in blood bags with anticoagulant. The new

donors in age group 21 to 44 years are in the high-risk category[14,18] and their blood is collected in blood packs without an anticoagulant as part of risk management for HIV and other transfusion-transmissible infections (TTIs).[19-22] As an additional safety measure for window period–phased donations there are instances when a dry pack is used on repeat donors. This applies to all lapsed repeat donors and repeat donors aged 21 to 30 years coming for their second donations. Furthermore, in cases where a high-HIV-risk repeat blood donation is suspected due to individual donor interview the donor is deferred from donation or a dry pack may be used. In some instances the dry packs are also used for other purposes other than HIV risk, such as malaria temporary deferrals. Donors with high risk for malaria will be temporarily deferred. After this period of temporary deferral, donors will be regarded as first-time donors. As such, the first donation will be in a dry pack. It is important to note that in all cases, the fulfillment of set established donor eligibility is critical. If the donor interview indicates high risk the donor is permanently deferred and no dry pack is used.

The option of predonation screening is not encouraged due to concerns that this may be associated with voluntary counseling and testing services thereby encouraging test seeking behavior (personal communication). We are unaware of any published evidence on this. This is a potential area for future research. It may be worth investigating the views of potential donors on these crucial aspects so that the service stands informed accordingly. The current practice is based on the notion that the experience of donating a full unit will help the donor to overcome donation fears and help the staff to identify donors at risk of adverse reactions. Donors are educated at the point of donation on the different blood pack types and blood use as indicated on the Donor Bill of Rights posted at all NBSZ donation facilities.

Tracking first-time and repeat donations

The type of pack used at each donation is identified and tracked at the point of blood collection and entered into the NBSZ nationally centralized electronic database introduced in 1994. The national electronic data management system allows all donations collected in dry and wet packs to be tracked and these are linked to the TTIs testing results and other demographic factors such as age. In this study units collected in dry packs are risk classified as first-time donations and those in wet packs as repeat donations based on the donor risk profile described above. We have used the term first-time conservatively to denote high-risk donations typical of adult new donors hence the use of the blood bag without an anticoagulant. The term new donor or donations is reserved for all donations where there is no previous donation. HIV reactivity data in first-time and repeat donations from 2002 to 2010 was retrieved and analyzed from the NBSZ national donor database. All the five NBSZ centers located in major cities (Harare, Bulawayo, Gweru, Mutare, and Masvingo) were nationally connected to the national database through wide area network by 2002. Consequently, the national database is representative of all centers as from 2002; hence this study includes all data from 2002 to 2010.

Serologic screening

From 2002 to 2006, an HIV antibody test was in use. As from 2007 the primary HIV test is the Architect HIV Ag/Ab Combo assay (Abbott Diagnostics, Wiesbaden, Germany), which is a chemiluminescent microparticle immunoassay. The chemiluminescent microparticle immunoassay detects HIV p24 antigen and antibodies to HIV-1 and HIV-2. If the primary test is nonreactive for HIV and other markers the unit is made available for distribution except if it is a first-time donation as per the donor risk classification. All reactive or discrepant samples from the primary test are repeated in duplicates using the same technology. If both duplicates are negative, the blood will be released; otherwise the samples and blood bags are further tested using an alternative technology (AxSYM HIV Ag/Ab Combo; Abbott Diagnostics) for confirming the test results. This assay is a microparticle enzyme immunoassay for the simultaneous detection of antibodies to HIV-1 and HIV-2 and HIV p24 antigen.

Transmission risks

HIV transmission risk of first-time donation was estimated from the prevalence (expressed as a proportion), window period (22 days, 2002-2006; 17 days, 2007-2010),[23] and duration in HIV Stage 1 and 2 (5 years, expressed in days) using the method published by van Hulst et al.[24]

For repeat donations, the residual risk of HIV transmission was estimated from the interdonation interval, incidence rate, and the window period (22 days, 2002-2006; 17 days, 2007-2010 expressed as a fraction of one year)[23] as described by Schreiber and colleagues.[25] To determine the incidence rates, seroconvertors (numerator) were identified on repeat donors who had a previous seronegative donation before the index donation during the year of follow-up. The interdonation intervals in days were obtained by obtaining the differences between all donations. These were summed up to give the denominator (person-years of observation) used in repeat donors risk calculation. For seroconverting donors, a correction factor, the preseroconversion interval midpoint, was used assuming that seroconversion occurred halfway during that period as proposed by Schreiber and colleagues.[25] HIV-positive repeat donors with a negative previous donation were defined as seroconvertors. Only repeat donors

with at least two donations within the follow-up period were considered. The window periods of 22 days (2002-2007) and 17 days (2007-2010) were decided based on the literature review[23,25] and review of the previous and current testing assays user package inserts. The residual risk was obtained by multiplying the window period and the incidence rates.

Statistical analysis

The HIV prevalence was calculated and the chi-square test was used to determine whether there was a significant difference between the HIV prevalence in first-time versus repeat donations. The relative risk (RR) was calculated and 95% CI for the RR determined. A p value of less than 0.05 was used to determine a significant difference. Trend analysis was done using Cochran-Armitage trend test. Residual risk estimates are reported.

RESULTS

From 2002 to 2010 a total of 116,058 (21%) and 434,695 (79%) donations were collected from first-time and repeat donors, respectively (Table 1). In first-time donors, the mean annual number of blood donation units was 12,895 (range, 9094-17,248 units) and in repeat donations it was 48,299 units (range, 27,293-58,682 units). These donations were all from voluntary nonremunerated donors and on average the numbers of annual blood donors are 10,001 (range, 6977-14,004) and 33,769 (range, 19,814-40,861) for first-time and repeat donors respectively. The number of first-time blood donations remained stable (linear trend $p = 0.726$) from 2002 to 2010. There was a significant overall continued downward trend (linear trend $p = 0.013$) in repeat blood donations which decreased by 53% in 8 years from 2002 (58,642) to 27,293 (2009) (Fig. 1). In 2010, there was a 74% increase in repeat blood donations.

Over the 9 years the overall HIV prevalence estimates are 1.29% (range, 0.89%-1.88%) and 0.42% (range, 0.27%-0.53%) for first-time and repeat donations, respectively. The overall RR was 3.1 (95% CI, 2.9-3.3; $p < 0.0001$) with RR ranging from 1.9 to 4 (Table 1). Thus, the RR of HIV positivity was 3.1 in blood donations from first-time relative to blood donations from repeat donors. Between 2002 and 2009, there was a significant decreasing trend in HIV seroprevalence (Cochran-Armitage test for linear trend, $\chi^2 = 37.01$, $p < 0.0001$) in first-time donors. When we added the 2010 data, which show a sharp upward trend in HIV prevalence from 0.95% (2009) to 1.88% (2010), the linear trend was marginally significant (Cochran-Armitage test for linear trend, $\chi^2 = 4.20$, $p = 0.04$). In repeat donations from 2002 to 2010 there was no significant linear trend in HIV prevalence (Cochran-Armitage test for linear trend, $\chi^2 = 0.14$, $p = 0.7082$). The trends in HIV are shown in Fig. 2.

From 2002 to 2010, the mean residual risk for HIV transmission in first-time donations was 1:7384 (range, 1:11,308-1:5356). In 2002, the highest residual risk was recorded as 1:5356 and in 2010 the risk was 1:5714 and this corresponded to the highest HIV prevalence recorded during the same year. However, this risk was averted, as these units are deliberately not channeled toward blood supply system. In the repeat blood donations, the mean residual risk of transmission was 1:5496 (range, 1:9943-1:3347). The highest risk of 1:3347 recorded in 2009 was the same period when the lowest number of repeat collections of 27,293 (65%) was recorded. On average the residual risk in first-time donations was 0.8 times (range, 0.3-1.3) lower than the residual risk estimated in repeat donations. Table 2 and Fig. 3 show the data on the HIV residual risk estimates.

When we further analyzed (data not presented) and compared the residual risk estimates in repeat donors in wet and dry packs we noted that there was a high residual risk in repeat donors who were dry packed (mean, 1:2538 vs. repeat, wet packs 1:5496 and first-time, dry packs 1:7384). We also further analyzed the overall data for all new donors (n = 242,739, 44% of total donations) by age groups irrespective of the pack type used. The overall residual transmission risk in the age groups 16 to 20 (n = 213,160), 21 to 29 (n = 20,998), 30 to 39 (n = 5343), 40 to 44 (n = 1246), and above 44 years (n = 1992) were 1:16,068; 1:4758; 1:2143; 1:2242, and 1:4972, respectively.

TABLE 1. HIV prevalence (national) in first-time and repeat donations, 2002 to 2010

	First-time (dry pack)		Repeat (wet pack)		
Year	Donations	HIV %	Donations	HIV %	RR (95% CI)
2002	14,365	1.55	58,642	0.43	3.6 (3.0-4.3)*
2003	11,792	1.42	57,301	0.39	3.6 (3.0-4.4)*
2004	12,440	1.41	55,343	0.42	3.4 (2.8-4.1)*
2005	12,629	1.53	58,074	0.53	2.9 (2.4-3.5)*
2006	17,248	0.89	51,262	0.41	2.2 (1.8-2.7)*
2007	9,094	1.00	42,872	0.27	3.7 (2.8-4.8)*
2008	12,477	1.14	36,347	0.37	3.1 (2.5-3.9)*
2009	14,661	0.95	27,293	0.51	1.9 (1.5-2.4)*
2010	11,352	1.88	47,561	0.47	4.0 (3.3-4.8)*
Total 2002-2010	116,058	1.29	434,695	0.42	3.1 (2.9-3.3)*

* $p < 0.0001$.

DISCUSSION

On blood collections it was noted that in 2010, there was a dramatic 74% increase in repeat blood donations and possible explanation for this is that Zimbabwe was recovering from the 2008 and 2009 worst operational era. Consequently, NBSZ was targeting to increase its

repeat blood collections, which had reduced to 27,293 in 2009. This was aimed at ensuring that there will be sufficient usable blood to fulfill the anticipated increased blood demand for maternity supported programs.

In the study period, the overall HIV prevalence in blood donations from first-time donors was significantly higher (1.29%) than the HIV in blood donations from repeat donors (0.42%). This is indicative of the usefulness of the policy of using dry packs in the HIV risk management in Zimbabwe. In both first-time and repeat blood donations, the HIV seroprevalence is lower than that in the general population,[3,4] which indicates the efficacy of donor motivation and selection procedures by NBSZ. The reasons for the HIV upsurge in 2010 in dry packs cannot be discerned now and is an area for further investigations.

Over the 2002 to 2010 period, the overall residual risk estimate in first-time donations was 1:7384 and in repeat donations it was 1:5496. The data seem to suggest that the residual risk estimates in first-time and repeat donations are comparable. This finding is consistent with the findings of Shang and coworkers,[23] who find a lower residual risk in first-time among Shenzhen blood donors, which is quite in contrast with Western blood donor populations. They speculated that this might be due to differing donor demographics and donation motivations in the population they studied. In Zimbabwe, just like Shenzhen the donor population is relatively young with 70% of donations coming from donors aged 16 to 20 years, which differ markedly with, say, Australia with median donor age of 44 years.[23] The young donors are generally considered highly mobile and prone to risky behaviors that may have the consequence of equilibrating the risk profile of first-time and repeat donors. As shown in Fig. 3, between 2006 and 2009 there was notable elevated residual risks in repeat donors over this period, which corresponds to the Zimbabwean harsh socioeconomic environment, which was also affecting donor recruitment and blood collections. This may have resulted in some donors ordinarily regarded as high risk having their blood collected in wet packs in an effort to ensure blood adequacy, hence contributing to the upward residual risk observed. The use of dry packs for other reasons other than HIV risk may also have contributed to the dilution effect. As argued by Shang and colleagues[23] it may as well be reflective of the fact that repeat donors in developing countries such as Zimbabwe fail to deliver the premium of "lowered risk" that is found in Western donor populations. These findings have implications for NBSZ to consider other risk management strategies including considerations for implementing more sensitive tests such as nucleic acid testing (NAT) so as to further safeguard the blood supplies in Zimbabwe.

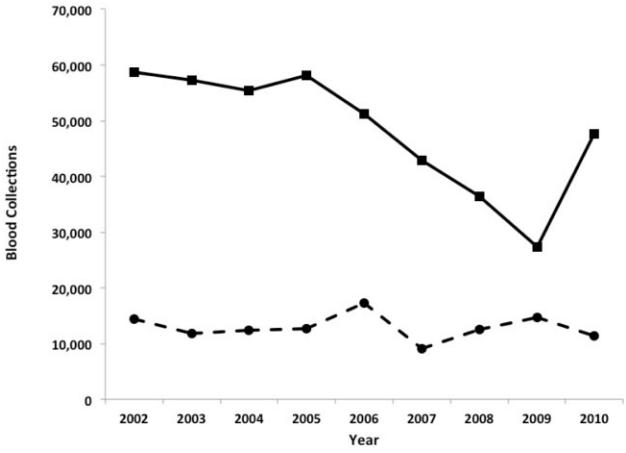


Fig. 1. Blood collection (national) trends in first-time (dry pack, –●–) and repeat (wet pack, –■–) donations, 2002 to 2010.

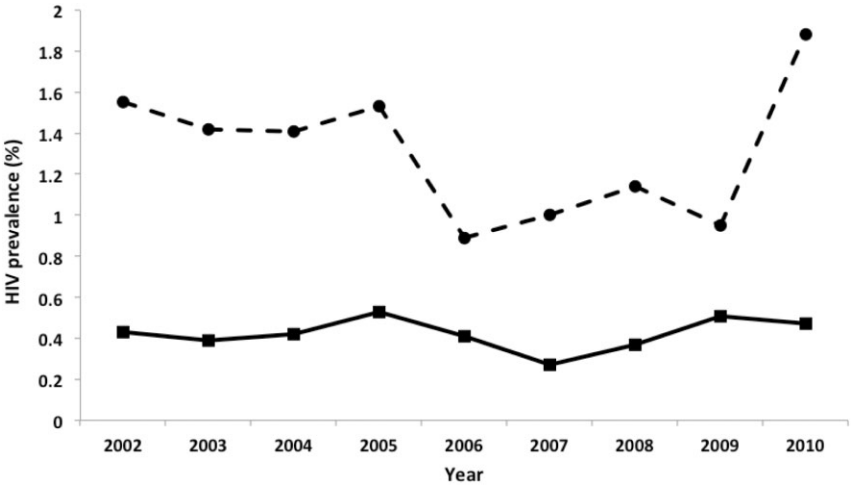


Fig. 2. HIV trends (national) in first-time (dry pack, –●–) and repeat (wet pack, –■–) donations, 2002 to 2010.

An independent report (R.Y. Dodd, S.E. Lloyd, R.K. Shamu, J. Emmanuel, D.A. Mvere, "Management of risk of HIV transmission by blood transfusion in Zimbabwe," 1995-96) on the study conducted among blood donors in Zimbabwe between July 1, 1993, and June 31, 1995, reported the HIV prevalence and incidence rates of 9.6% (95% CI, 9.2%-10.0%) and 0.56% (95% CI, 0.51%-0.61%), respectively. The residual risk of infection from blood supply was reported as 1:3670 (1:4268-1:3157). The percentage reduction of prevalence from 9.6% (1993-1995) to 1.29% (2002-2010) in dry packs is quite remarkable. The incidence rate reduced from 0.59% to 0.42% (crude rate, Table 1) over the same period. The residual risk in 1993 to 1995 of 1:3650 is higher than the current estimated residual risk of 1:5496 (repeat donations) in total of 2002 to 2010. This

TABLE 2. Residual risk estimates (per 100,000 donations) in first-time (dry pack) and repeat (wet pack) donations (national), 2002 to 2010

	Repeat donors (wet packs)					First-time donors (dry packs)		
Year	Number of seroconversions	Number of person-years	Incidence rate (per 100,000 person-years)	Residual risk* (per 100,000 donations)	Risk (rate of infections donations)	HIV prevalence (%)	Residual risk† (per 100,000 donations)	Risk (rate of infections donations)
2002	21	7,070.9	297.0	17.9	1:5,590	1.55	18.7	1:5,356
2003	23	6,584.4	349.3	21.0	1:4,753	1.42	17.1	1:5,846
2004	18	6,414.4	280.6	16.9	1:5,916	1.41	17.0	1:5,887
2005	23	6,921.9	332.3	20.0	1:4,996	1.53	18.4	1:5,426
2006	10	5,988.7	167.0	10.1	1:9,943	0.89	10.7	1:9,327
2007	15	4,561.2	328.9	15.3	1:6,533	1.00	9.3	1:10,743
2008	19	3,651.3	520.4	24.2	1:4,129	1.14	10.6	1:9,423
2009	17	2,648.4	641.9	29.9	1:3,347	0.95	8.8	1:11,308
2010	16	5,442.9	294.0	13.7	1:7,309	1.88	17.5	1:5,714
2002-2006	95	32,980	288.1	17.4	1:5,764	1.36	16.4	1:6,104
2007-2010	67	16,304	410.9	19.1	1:5,228	1.24	11.5	1:8,663

* Calculated based on method published by Schreiber and colleagues[25]; window period of 22 days (2002-2006; HIV antibody) and 17 days (2007-2010; HIV combo) was used based on literature review[23,25] and the past and current assay in use at NBSZ. Residual risk is obtained by multiplying the window period expressed as a fraction of year (365.25) with the incident rate.

† Calculated based on method published by van Hulst and colleagues[24]; the residual risk is obtained by multiplying the prevalence as a proportion with window period (22 days, 2002-2006; 17 days, 2007-2010)[23,25] divided by duration WHO HIV Stages 1 and 2 (in days) estimated as 5 years.

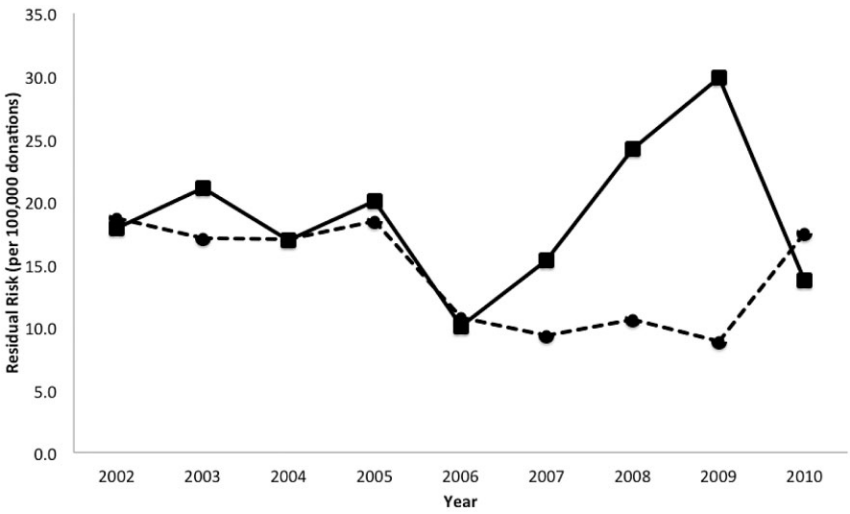


Fig. 3. HIV residual risk (per 100,000 donations) trends in first-time (dry pack, –●–) and repeat (wet pack, —■—) donations (national), 2002 to 2010.

reflects positive gains in reducing residual risk by NBSZ under difficult circumstances and this provides the much-needed confidence in the blood supply. The selected residual risk model by Schreiber and coworkers[25] used in this study for the repeat donors requires NAT screening, which is currently not performed in Zimbabwe. There are also other available published residual risk models for first-time donors[23] and these may need to be considered in further refining the provided residual risk estimates in Zimbabwe. It is recommended that sensitivity analysis of residual risk estimation models be applied to blood donor data in Zimbabwe. The reported residual risk estimates need to be cautiously interpreted in light of these limitations.

We note the apparent differences in the conclusions based on prevalence and residual transmission risk in first-time and repeat donors (Figs. 2 and 3). While it is quite evident from the prevalence data that the dry pack HIV prevalence is always above the HIV prevalence in repeat donors (RR, 3.1; $p < 0.0001$), a different pattern is exhibited for the residual transmission risk especially for the period 2006 to 2009 for the reasons postulated above. It can be concluded that in addition to monitoring prevalence data, the residual transmission risk data are also vital to ensure a balanced decision-making process. The high residual risk estimates in repeat donors in dry packs (1:2538 vs. repeat wet packs 1:5496) seems to suggest that when appropriately applied, the dry pack policy is an effective risk management strategy. This is further supported by the fact that there is a sevenfold increase of residual risk in new donors in the age groups 30 to 39 years (1:2143) and 40 to 44 years (1:2242) compared to the age group 16 to 20 years (1:16,068). In the age groups 20 to 29 years (1:4758) and above 44 years (1:4972) we observed a threefold increase in residual risk compared to the new donors in the age group 16 to 20 years. The comparable residual risks observed in the age group 20 to 29 and older than 44 years were not expected and the dry pack policy may need to be reviewed in line with these

results. The stratified analysis by age groups provides evidence of the need to review the application of different age specific policies to new donors. However, we note that 88% of all the overall new donors are in the age group 16 to 20 years and this should be considered when interpreting these results. Although the NBSZ developed its window period risk algorithm in 1994, its use to evaluate key policies such as dry pack policy has been limited although the overall window period risk estimates are routinely reported in its annual reports.

In a high-HIV-prevalence setting such as Zimbabwe the use of appropriate donor selection methods complemented with advanced testing technologies will improve blood safety. In South Africa where the HIV prevalence in the general population is 11.4%,[26] the South Africa National Blood Service reported their residual risks in first-time and repeat donors as 1:25,641 (3.9/100,000 donations) and 1:40,000 (2.5/100,000 donations), respectively, based on study by Heyns and coworkers.[26] After individual-donation NAT introduction in 2005, the South Africa National Blood Service reported[27] that the residual transmission risk of window phase donations was estimated at 1:479,000, which indicates enhanced blood safety through implementation of an advanced testing technology. In a residual transmission risk of TTIs study in sub-Saharan Africa by Jayaraman and colleagues[28] they estimated that the risk of HIV transmission is 1:1000 donations (one infection per 1000 units), which is high compared to the Zimbabwean data. This may be indicative of different risk management strategies in the sub-Saharan blood services.

The observed significant HIV declining trend in first-time donors and no significant trend in repeat donors portrays a mixed message and this needs to be properly examined. Zimbabwe, like most developing countries, has generalized HIV epidemic and the relationship between general and donor population HIV needs proper monitoring and evaluation as this impacts the blood safety strategies. It is generally accepted that the HIV prevalence in blood donors if cautiously interpreted can provide useful insight to the epidemic condition in the general population and vice versa.[29] We are currently conducting such studies in Zimbabwe, which have been undertaken in other settings.[30,31]

NBSZ through its strengthened blood safety program is aware of its potential contribution to the continued fight against HIV and AIDS in Zimbabwe. The WHO estimates that up to 3% of HIV infections worldwide are transmitted through the transfusion of contaminated blood and blood products.[32] However, to date NBSZ does not have documented cases of HIV transmission through blood transfusion in Zimbabwe. However, we note that the reporting structures for such cases may still be underdeveloped in Zimbabwe. This is an area that may still need further attention in Zimbabwe. One of the key recommended strategies to prevent transfusion-associated HIV transmission is the appropriate use of blood products.[18] Zimbabwe developed and published guidelines for prescribing blood,[33] which are used to educate and guide clinicians on appropriate use of blood and blood products.

The significantly high HIV prevalences recorded in first-time over repeat donations seems to suggest the effectiveness of the HIV risk management strategy for HIV through screening and discarding first-time donations in Zimbabwe. In settings where there is high HIV prevalence and other TTIs in the general population, the use of dry pack policy or alternative strategies such as test-tube samples may be considered as a useful risk management strategy in blood transfusion. However, this should be considered in light of residual risk estimates, which in this study indicate comparable residual risk in first-time and repeat donors. We are of the view that before policies can be changed further research is warranted. The findings presented in our article justify investing resources in more precise estimation of the residual risk between first-time and repeat donors and a rigorous cost-effectiveness evaluation. Furthermore, there remains uncertainty surrounding the numbers of undetected TTIs due to the limited capacity for hemovigilance. Additionally, the difference in risk of transmission of TTI is unknown between first-time and repeat donors. It is also important to ensure that the review of blood safety strategies including the application of the dry pack policy needs to be balanced with the availability of adequate and constant blood supplies. The policy of collecting dry pack from first-time donors has evolved over years; this article opens the discussion whether it should be continued or changed. It is perceived that collecting tubes could attract high-risk donors by suggesting a voluntary counseling and testing service and there is potential risk of stigmatization especially in a school donor population. Further research could compare the risk profile, incidence, and prevalence of collecting tube versus dry packs in first-time or prospect donors. Yet, given the potential wastage of valuable resources, future studies should focus on the cost-effectiveness and utility of screening and discarding first-time donations[34] in Zimbabwe.

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CONFLICT OF INTEREST

The authors declare that they have no conflicts of interest relevant to the manuscript submitted to **TRANSFUSION.**

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